

Call Center Intel — Agentic AI for Sales & Service Conversations

Industry Context

Industry: built on an Insurance & Healthcare application, but not tied to it — it fits any contact centre whose agents speak directly to prospects or members.

Trends: speech-to-text and language models are now cheap and fast enough to run on every call rather than a sampled review, while regulators tighten what may be said on a sales call.

Compliance exposure: violations surface only when a transcript is read, days or weeks later — by then the cost is remediation, penalties and reputational damage.

Copilot, not chatbot: the AI works symbiotically with the application — reading live data, driving the screens, filling records and placing calls for the agent.

Use case: a live AI layer that assists the agent while they work, and flags a breach while they are still speaking.

Client's business problems

Compliance after the fact. By the time a transcript is read, the words are said and the exposure is booked.

QA from a sample. A handful of calls per agent are reviewed; the standard lives in each reviewer's head, so scoring drifts.

Ramp-up on live prospects. A new agent's first hard call is a real one, with a real compliance obligation.

Missed calls become lost leads. Prospects dial when every agent is busy or off-shift, and the lead dies on a twenty-second ring.

The screen competes with the customer. Agents type through the call instead of listening — and an assistant that only pushes text back adds one more thing to read.

AI cost can't be traced. Spend arrives as one vendor invoice — no per-call or per-agent figure to manage against.

Functional Overview

The Solution: Call Center Intel adds a real-time AI layer over the sales and service conversation itself — four pillars on one pipeline. (1) An **Agentic Copilot** with 23 governed tools that answers from live business data, drives the UI, drafts emails and record updates for one-click approval, and places calls from natural language. (2) **Real-Time Conversation Intelligence** — live transcription, two-layer compliance (deterministic rules in under a second, plus an AI layer that catches paraphrased violations), emotion-shift detection and proactive coaching. (3) A **Supervisor Command Centre & AI Governance** where business users edit compliance rules, coaching rules and QA scoring criteria with no deployment. (4) **Cost Transparency** — per-call and per-agent AI cost from real usage logs, split by feature and drillable to the transcript behind any figure. A voice agent also answers inbound calls on hand-over, ring-timeout or out-of-hours — disclosing that it is automated, building the customer record mid-call and booking the callback.

How it works: call audio is transcribed live with speaker separation; one merged model call returns emotion, coaching and compliance together, keeping cost and latency predictable; hybrid retrieval grounds every answer in live plan, provider, pharmacy and CRM data so the model never guesses; results stream to the agent and supervisor screens as the conversation happens. Every AI interaction is logged for audit and cost attribution.

Fit with EXL sales: EXL consults on, builds and runs systems for some of the largest enterprises in the world. This is a complete end-to-end implementation, not a concept deck — it can be demonstrated as-is to any contact centre client from the first meeting of a pursuit. That makes it both a door-opener for new business and the clearest proof point of EXL's AI capability we can put in front of a client.

Technological Highlights

Core technologies: LangGraph tool-using agent (23 governed tools); Claude via OpenRouter for reasoning; Deepgram streaming speech-to-text; nomic-embed-text embeddings driving hybrid keyword + vector retrieval. Qdrant vector database for retrieval; MongoDB for application and reference data. Twilio programmable voice with bidirectional media streams; Cloudflare tunnel and Caddy gateway; containerised and cloud-portable. Node 20 + Express 5 (TypeScript) API, with a React 19 / TypeScript single-page front end.

Innovation factor: compliance and coaching enforced **during** the call, not after it; two-layer detection (instant rules + AI for paraphrase) so nothing is traded off between speed and nuance; no-code AI governance — business users, not engineers, own the rules and QA criteria, effective on the very next call.

Scalability: cost-engineered by design — one merged structured LLM call per 15-second window, prompt caching and typed outputs keep cost per call predictable as volume grows. Stateless containerised API scales horizontally; per-call and per-user cost telemetry projects spend to a 100-agent workforce from real usage logs.

Integration: REST + SSE surfaces and a governed tool layer sit over existing CRM, telephony and master data — the AI calls the same tools whether it is on chat or voice. Identity is injected server-side, never chosen by the model; a disabled capability is withheld from the AI rather than discouraged in a prompt.

Business Impact

Revenue streams: the primary stream is expansion — open new lines of business with organisations whose core operation is the contact centre, use this as living proof of what AI can do for them, and implement across the rest of their operation from there. Commercially it lands as a per-agent-seat subscription for the copilot and conversation intelligence, per-call pricing for automated inbound handling, and an outcome-linked QA and compliance tier — each also strengthening managed-operations deals.

Customer / stakeholder value: agents get their hours back for people ready to talk; supervisors are effectively present on every live call rather than a sampled handful; compliance teams get evidence-quoted scorecards on 100% of conversations with a full audit trail; CFOs get per-call cost attribution instead of a single opaque invoice; and business owners can change the rules themselves without a release cycle.

Potential impact: captured leads that would otherwise die unanswered on ring-out and out-of-hours; QA coverage moves from a reviewed sample to every call; violations are caught in the moment rather than weeks later, reducing remediation and regulatory exposure; new-agent ramp shortens because coaching and hands-free data capture arrive from day one; supervisor span of control widens without added headcount.

Cost structure: dominant variable cost is LLM and STT usage per call minute — already measured and attributable in the platform; fixed costs are container hosting, Qdrant and MongoDB. The POC runs today on commodity cloud infrastructure with open-source components, so the build investment is incremental engineering rather than new licensing.